

PART-A

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INSTRUCTIONS

1. Check that you have **10 printed pages**. The PART-A of the examination has **3 questions**, for a total of **20 points**.
2. Attempt **all questions**.
3. There is **no credit** for a solution if the appropriate work is not shown, even if the answer is correct.
4. Write your answers in this booklet and only in the **space marked for answers**. Any answer written in any space other than the designated space **will not be evaluated**.
5. The pages numbered **8 to 10** are kept as **additional pages**. If you cannot able to fit a complete answer in the space provided just after the question, you can use these three pages by referring the page number of additional page in the original space.
6. At the beginning of the examination, a supplementary sheet has been provided only for rough work. Should you need more, please ask the invigilator.
7. No question requires any clarification from the instructors or invigilators, even if a question has an error or incomplete data. The students are advised to write answer according to their understanding or write reasons for why it is not possible to solve partially or completely that question by citing errors/insufficient data.

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Question:	1	2	3	Total
Points:	5	6	9	20
Score:	2	2	9	13

QUESTIONS

1. (5 points) Let Y and ϵ are the response variable and the error in a regression problem, respectively. θ_1 and θ_2 are the two parameters. Let $y_i, i = 1, 2, 3$, be the observed values of the response variable. Consider,

$$Y_1 = \theta_1 + \theta_2 + \epsilon_1,$$

$$Y_2 = \theta_1 - 2\theta_2 + \epsilon_2,$$

$$Y_3 = 2\theta_1 - \theta_2 + \epsilon_3,$$

where $E(\epsilon_i) = 0$, and $V(\epsilon_i) = \sigma^2, i = 1, 2, 3; \sigma > 0$. Find the least square estimates (LSE) of θ_1 and θ_2 .

$\Rightarrow$ We need to minimize the function

$$\sum_{i=1}^3 \epsilon_i^2$$

$$= \epsilon_1^2 + \epsilon_2^2 + \epsilon_3^2$$

$$\text{or, } Q(\theta_1, \theta_2) = (y_1 - \theta_1 - \theta_2)^2 + (y_2 - \theta_1 + 2\theta_2)^2 + (y_3 - 2\theta_1 + \theta_2)^2$$

~~First~~ Taking the derivatives w.r.t. θ_1 & θ_2 and equating to 0, we get,

$$Q_{\theta_1}(\theta_1, \theta_2) = -2(y_1 - \theta_1 - \theta_2) - 2(y_2 - \theta_1 + 2\theta_2) - 4(y_3 - 2\theta_1 + \theta_2) = 0$$

$$\Rightarrow y_1 + y_2 + 2y_3 - \theta_1 - \theta_2 - \theta_1 + 2\theta_2 - 4\theta_1 + 2\theta_2 = 0$$

$$\Rightarrow 2\theta_1 + 3\theta_2 = -(y_1 + y_2 + 2y_3) \quad \text{--- (1)}$$

$$Q_{\theta_2}(\theta_1, \theta_2) = -2(y_1 - \theta_1 - \theta_2) + 4(y_2 - \theta_1 + 2\theta_2) + 2(y_3 - 2\theta_1 + \theta_2) = 0$$

$$\Rightarrow y_1 - 2y_2 - y_3 - \theta_1 - \theta_2 + 2\theta_1 - 4\theta_2 + 2\theta_1 - \theta_2 = 0$$

$$\Rightarrow 3\theta_1 - 6\theta_2 = -(y_1 - 2y_2 - y_3) \quad \text{--- (2)}$$

~~Solving (1) & (2), we get~~

~~$$\theta_1 = -\frac{3}{7}(y_1 + y_3)$$~~

~~Solving~~

Multiply (1) by 2 and add to (2):

$$7\theta_1 = -2(y_1 + y_2 + 2y_3) - (y_1 - 2y_2 - y_3)$$

$$= -3y_1 - 3y_3$$

$$\Rightarrow \theta_1 = -\frac{3}{7}(y_1 + y_3)$$

Putting this value of θ_1 in (1):

$$3\theta_2 = -(y_1 + y_2 + 2y_3) + \frac{6}{7}(y_1 + y_3)$$

$$= -\frac{y_1}{7} - y_2 - \frac{8}{7}y_3$$

$$\Rightarrow \theta_2 = \cancel{0} - \frac{y_1}{21} - \frac{y_2}{3} - \frac{8y_3}{21}$$

$$= -\frac{1}{21}(y_1 + 7y_2 + 8y_3)$$

$\therefore$ the LSE's are:

$$\hat{\theta}_1 = -\frac{3}{7}(y_1 + y_3)$$

$$\hat{\theta}_2 = -\frac{1}{21}(y_1 + 7y_2 + 8y_3)$$

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2. Let U_1 and U_2 be two independent uniform random variables on the interval $(0, 1)$. Suppose that $\mathbf{X} = (X_1, X_2, X_3)'$, where $X_1 = U_1$, $X_2 = U_2$, and $X_3 = U_1 + U_2$.

(a) (3 points) Compute the correlation matrix ρ of $\mathbf{X}$.

(b) (3 points) Find the variance of first principal component based on the correlation matrix that you find in part (a).

$$\Rightarrow U_1, U_2 \stackrel{i.i.d.}{\sim} U(0, 1)$$

$$\text{PDF of } U_1: f(u_1) = 1 \quad \forall u_1 \in (0, 1)$$

$$\text{PDF of } U_2: f(u_2) = 1 \quad \forall u_2 \in (0, 1)$$

$$\therefore \text{JPDF of } U_1, U_2: f(u_1, u_2) = 1 \quad \forall u_1, u_2 \in (0, 1)$$

$$\text{Let } \begin{cases} W = U_1 + U_2 \\ V = U_1 - U_2 \end{cases} \Rightarrow \begin{cases} U_1 = \frac{W+V}{2} \\ U_2 = \frac{W-V}{2} \end{cases}$$

$$\text{Jacobian: } |J| = \begin{vmatrix} \frac{\partial U_1}{\partial W} & \frac{\partial U_1}{\partial V} \\ \frac{\partial U_2}{\partial W} & \frac{\partial U_2}{\partial V} \end{vmatrix} = \begin{vmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{vmatrix}$$

$$\therefore f_{W,V}(w, v) = 1 \cdot \frac{1}{2} = \frac{1}{2} \quad \forall w, v \in (0, 1)$$

$$= \left| -\frac{1}{4} - \frac{1}{4} \right| = \frac{1}{2}$$

Marginal PDF of W :

$$f(w) = \int_0^1 \frac{1}{2} dv = \frac{1}{2}$$

$$\mathbb{E}[W] = \int_0^1 w \cdot \frac{1}{2} dw = \frac{1}{4}$$

$$\text{Var}(W) = \mathbb{E}[W^2] - (\mathbb{E}[W])^2$$

$$= \int_0^1 w^2 \cdot \frac{1}{2} dw - \left(\frac{1}{4}\right)^2 = \frac{1}{6} - \frac{1}{16} = \frac{5}{48}$$

Given, $\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix}$, $\therefore \mathbf{X} = \mathbb{E}[\mathbf{X}] = \begin{bmatrix} \mathbb{E}[U_1] \\ \mathbb{E}[U_2] \\ \mathbb{E}[U_1 + U_2] \end{bmatrix}$

or, $\mu = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{4} \end{bmatrix}$

For uniform ~~R.V.~~ R.V.,

$$E[X] = \frac{a+b}{2} \quad \text{and} \quad \text{Var}(X) = \frac{(b-a)^2}{12}$$

Here for X_1 and X_2 , $a=0$ and $b=1$.

$$\therefore E[X_1] = E[X_2] = \frac{1}{2}$$

$$\text{Var}(X_1) = \text{Var}(X_2) = \frac{1}{12}$$

Since X_1 and X_2 are independent,

$$\therefore \text{Corr}(X_1, X_2) = 0$$

$$\begin{aligned} \text{Cov}(X_1, X_3) &= \text{Cov}(X_1, X_1 + X_2) \\ &= \text{Var}(X_1) + \text{Cov}(X_1, X_2) \\ &= \frac{1}{12} \end{aligned}$$

$$\Rightarrow \text{Corr}(X_1, X_3) = \frac{\frac{1}{12}}{\sqrt{\frac{1}{12}} \cdot \sqrt{\frac{5}{48}}} = \frac{\frac{1}{12} \times \sqrt{12} \times \frac{\sqrt{48}}{5}}{1} = \frac{2}{5}$$

$$\text{similarly, } \text{Corr}(X_2, X_3) = \frac{2}{5}$$

$$\text{and } \text{Corr}(X_i, X_i) = 1 \quad \forall i = 1, 2, 3$$

$\therefore$ Correlation matrix of $X = (X_1, X_2, X_3)^T$ is:

$$\rho_{3 \times 3} = \begin{matrix} & \begin{matrix} X_1 & X_2 & X_3 \end{matrix} \\ \begin{matrix} X_1 \\ X_2 \\ X_3 \end{matrix} & \begin{bmatrix} 1 & 0 & \frac{2}{5} \\ 0 & 1 & \frac{2}{5} \\ \frac{2}{5} & \frac{2}{5} & 1 \end{bmatrix} \end{matrix}$$

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* Part (b) continued on additional page (9).

3. Let the matrix of distance be

$$\begin{array}{c} \begin{matrix} & 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} \begin{bmatrix} 0 & & & \\ 1 & 0 & & \\ 11 & 2 & 0 & \\ 5 & 3 & 3.5 & 0 \end{bmatrix} \end{array}$$

- (a) (6 points) Cluster the four items using each of the following procedures:
- Complete linkage hierarchical procedure.
 - Average linkage hierarchical procedure.
- (b) (3 points) Draw the dendrograms and compare the results obtained using previous two procedures.

(a) (i) Iteration 1:

The minm. distance in the given matrix is

$d_{2,1} = 1$. So we cluster 1 & 2 and calculate the new distances by taking maximum (complete linkage).

$$d_{(12),3} = \max \{ d_{1,3}, d_{2,3} \} = \max \{ 11, 2 \} = 11$$

$$d_{(12),4} = \max \{ d_{1,4}, d_{2,4} \} = \max \{ 5, 3 \} = 5$$

Therefore, the new distance matrix is:

$$\begin{array}{c} \begin{matrix} (12) & 3 & 4 \end{matrix} \\ \begin{matrix} (12) \\ 3 \\ 4 \end{matrix} \begin{bmatrix} 0 & & \\ 11 & 0 & \\ 5 & \textcircled{3.5} & 0 \end{bmatrix} \end{array}$$

Iteration 2: The minm. distance is $d_{4,3} = 3.5$

So we cluster 3 & 4 and calculate new distances:

$$\begin{aligned} d_{(34),(12)} &= \max \{ d_{3,(12)}, d_{4,(12)} \} \\ &= \max \{ 11, 5 \} = 11 \end{aligned}$$

Hence, the new distance matrix is:

$$\begin{matrix} & (12) & (34) \\ (12) & 0 & \\ (34) & \textcircled{11} & 0 \end{matrix} \begin{bmatrix} \\ \\ \end{bmatrix}$$

~~The new~~

Iteration 3: The min. distance now is $d_{(34)(12)} = 11$.

So we club (34) & (12) together, which forms our final cluster.

$$\begin{matrix} & (1\ 2\ 3\ 4) \\ (1\ 2\ 3\ 4) & \begin{bmatrix} 0 \end{bmatrix} \end{matrix}$$

We had $n = 4$ items and we ran our algorithm for $n - 1 = 3$ iterations. Hence, we are done.

(ii) Now we proceed with average linkage.

Iteration 1: Minm. distance in given matrix is

$d_{2,1} = 1$. So we club 1 & 2 together.

We calculate the distances as:

$$d_{(12),3} = \frac{d_{1,3} + d_{2,3}}{2} = \frac{11 + 2}{2} = 6.5$$

$$d_{(12),4} = \frac{d_{1,4} + d_{2,4}}{2} = \frac{5 + 3}{2} = 4$$

So, the new matrix is:

$$\begin{matrix} & (12) & 3 & 4 \\ (12) & 0 & & \\ 3 & 6.5 & 0 & \\ 4 & 4 & \textcircled{3.5} & 0 \end{matrix} \begin{bmatrix} \\ \\ \\ \end{bmatrix}$$

ADDITIONAL PAGES FOR ANSWERS

Iteration 2: New our minimum is $d_{4,3} = 3.5$.

We club 3 & 4. Now, we calculate new distances as!

$$d_{(34), (12)} = \frac{d_{3,1} + d_{3,2} + d_{4,1} + d_{4,2}}{4}$$

$$= \frac{11 + 2 + 5 + 3}{4}$$

$$= 5.25$$

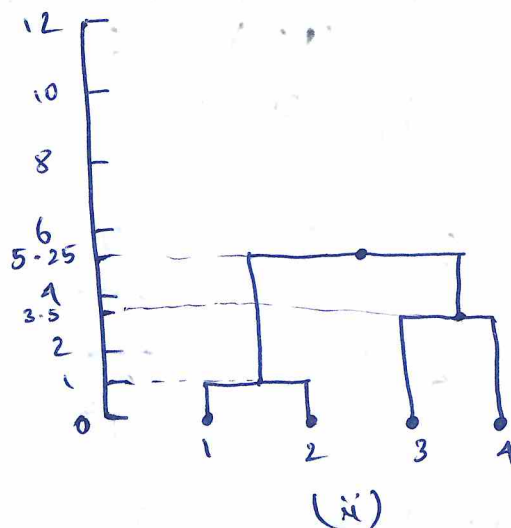
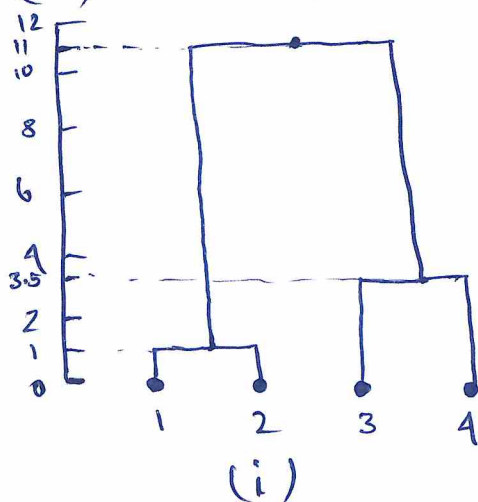
$$\begin{matrix} (12) & (34) \\ (12) & \begin{bmatrix} 0 & \\ & \end{bmatrix} \\ (34) & \begin{bmatrix} 5.25 & \\ & 0 \end{bmatrix} \end{matrix}$$

Iteration 3: Minm. distance is $d_{(34), (12)} = 5.25$. So we club (34) & (12) together to form one single cluster (1234).

$$\begin{matrix} (1234) \\ (1234) & \begin{bmatrix} 0 \end{bmatrix} \end{matrix}$$

Our algorithm has finished after $3 (= n-1)$ steps.

(b) Dendrograms: (Distance on y-axis and items on x-axis)



(i) is the dendrogram for complete linkage.

(ii) " " " " average " "

Observation: The distances for the first two iterations are same for both the procedures. But in the last iteration, the merging distance is much smaller ($= 5.25$) for the average linkage as compared to complete linkage ($= 11$).

* 2(b) We have,

$$P = \begin{bmatrix} 1 & 0 & \frac{2}{5} \\ 0 & 1 & \frac{2}{5} \\ \frac{2}{5} & \frac{2}{5} & 1 \end{bmatrix}$$

The variance of the first PC will be equal to the largest eigenvalue of P .

To find eigenvalues:

$$|P - \lambda I| = 0$$

$$\Rightarrow \begin{vmatrix} 1-\lambda & 0 & \frac{2}{5} \\ 0 & 1-\lambda & \frac{2}{5} \\ \frac{2}{5} & \frac{2}{5} & 1-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (1-\lambda) \left[(1-\lambda)^2 - \frac{4}{25} \right] + \frac{2}{5} \left[0 - \frac{2}{5} (1-\lambda) \right] = 0$$

$$\Rightarrow (1-\lambda)^3 - \frac{8}{25} (1-\lambda) = 0$$

$$\Rightarrow 1-\lambda = 0 \quad \text{or} \quad (1-\lambda)^2 = \frac{8}{25}$$

$$\Rightarrow \lambda = 1 \quad \text{or} \quad \lambda = 1 + \frac{2\sqrt{2}}{5} \quad \text{or} \quad \lambda = 1 - \frac{2\sqrt{2}}{5}$$

Clearly, $\lambda^* = 1 + \frac{2\sqrt{2}}{5}$ is the largest eigenvalue.
If Y_1 is the first PC, then

$$\text{Var}(Y_1) = \lambda^* = 1 + \frac{2\sqrt{2}}{5}$$

$$= 1 + 0.56569$$

$$= 1.56569$$

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